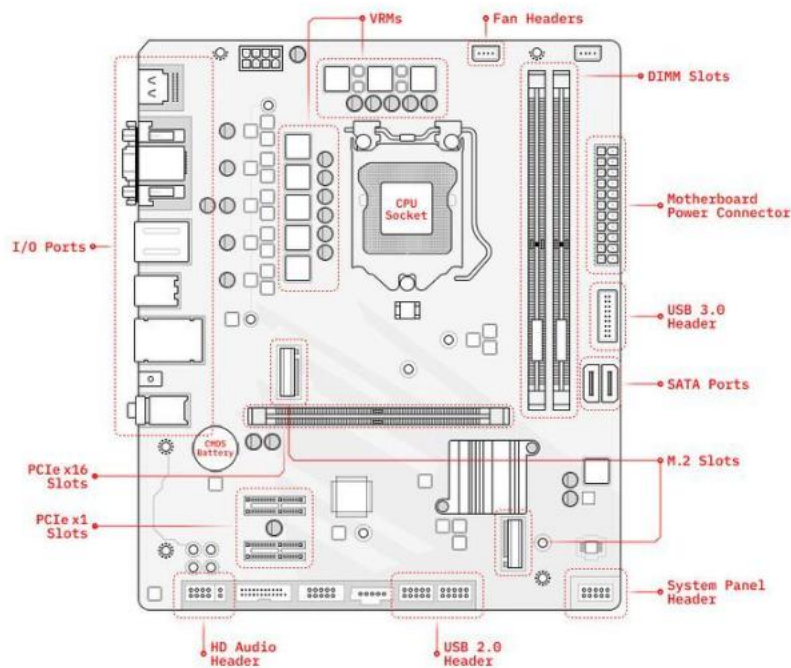


DAY 2- MOTHERBOARD AND CPU THEORY:

Tasks

1.

Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



2.

Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

A CPU socket is the physical connector on a motherboard where the processor is installed. It provides the electrical connection between the CPU and the motherboard. Different CPUs are designed for specific socket types, so a processor cannot simply be installed into any motherboard.

The socket, motherboard chipset, and BIOS must all support the processor. Three types of CPU sockets are:

1. LGA1151

LGA1151 is an Intel CPU socket used for several generations of Intel desktop processors. It has 1,151 contacts and was used with 6th, 7th, 8th and 9th generation Intel desktop CPUs, although different generations require compatible chipsets. An example of a processor using this socket is the Intel Core i7-8700.

2. AM4

AM4 is an AMD CPU socket designed for many AMD Ryzen desktop processors. It became popular because it supported several generations of Ryzen CPUs. For example, the AMD Ryzen 5 5600 uses the AM4 socket and is a 6-core, 12-thread desktop processor.

3. LGA1200

LGA1200 is an Intel socket used mainly with 10th and 11th generation Intel desktop processors. It has 1,200 contacts and replaced LGA1151 for these generations. An example of a compatible processor is the Intel Core i5-10600K.

3.

Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

Primary functions are:

CPU: The CPU is the main processor. It executes instructions, performs calculations, controls program

operations and coordinates data movement between hardware components.

Chipset: The chipset is the motherboard's supporting communication/control logic. It helps manage communication between the CPU and devices such as USB, storage, expansion slots and other peripherals, depending on the platform.

Opening Spotify: When you open Spotify, the CPU runs the operating-system and Spotify instructions. The motherboard platform/chipset and its connected controllers help move data between the CPU, RAM, storage and USB/network devices. RAM holds the program and active data, while the storage drive supplies the application files. Together, these components let Spotify load, connect to the network and play music smoothly.

4.

Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.

